



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# **LESSON PLAN**

For

## **AI Vibe Coding for Multi-Agents System**

TGS Ref No: TGS-2020503207

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 1.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                    | Author        |
|----------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1.0            | 11 August 2026            | Initial release — 2-day lesson plan for AI Vibe Coding for Multi-Agents System, aligned to the published course outline (Topics 1-5) and the 11 hands-on labs. | Dr Alfred Ang |

## TABLE OF CONTENTS

|                                                                      |          |
|----------------------------------------------------------------------|----------|
| <b>Course Information.....</b>                                       | <b>4</b> |
| <b>Learning Outcomes.....</b>                                        | <b>4</b> |
| <b>Skills Framework Alignment.....</b>                               | <b>4</b> |
| <b>Assessment.....</b>                                               | <b>4</b> |
| <b>Training Resources.....</b>                                       | <b>5</b> |
| <b>Course Schedule.....</b>                                          | <b>5</b> |
| Day 1 — Agent Foundations & Vibe Coding for Multi-Agent Systems..... | 5        |
| Day 2 — Agent SDKs, MCP, Sub-Agents & Assessment.....                | 6        |
| <b>Lab Reference (aligned to the course outline).....</b>            | <b>7</b> |
| <b>Learning Outcome to Lab Mapping.....</b>                          | <b>7</b> |

## Course Information

|                             |                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------|
| <b>Course Title</b>         | AI Vibe Coding for Multi-Agents System                                                      |
| <b>WSQ Course Reference</b> | TGS-2020503207                                                                              |
| <b>Training Provider</b>    | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                                          |
| <b>Skills Framework</b>     | Analytics and Computational Modelling (ICT-DIT-3001-1.1)                                    |
| <b>Duration</b>             | 2 days · 8 training hours per day (16 hours), plus 2 hours assessment                       |
| <b>Daily Timing</b>         | 9:30 am – 7:00 pm (8 instructional hours, plus a 1-hour lunch and two 15-minute tea breaks) |
| <b>Mode</b>                 | Instructor-led, hands-on multi-agent development labs in Python                             |
| <b>Trainer</b>              | Dr Alfred Ang                                                                               |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain the components of a modern AI agent — skills, memory, tools and the Model Context Protocol (MCP) — and the progression from single-agent to multi-agent collaboration.
- LO2: Apply a structured vibe coding workflow with context engineering to specify, generate and verify agent code reliably.
- LO3: Build a collaborative multi-agent system with the OpenAI Agents SDK using structured outputs, tool calling and supervisor routing, and deploy it with Streamlit.
- LO4: Build a collaborative multi-agent system with the Google Gemini Agent SDK and deploy it with Streamlit.
- LO5: Orchestrate tools through MCP and design hierarchical sub-agent architectures for delegated, specialised work.

## Skills Framework Alignment

TSC Title: Analytics and Computational Modelling | TSC Code: ICT-DIT-3001-1.1

- A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories
- A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements
- A3 - Utilise a range of statistical methods and analytics approaches to analyse data
- A4 - Evaluate the performance and suitability of models against business requirements

## Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on multi-agent build tasks, 1 hour, open book.

- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

## Training Resources

- Course slide deck (Trainer Slides / Learner Slides) — downloadable from the LMS/TMS portal.
- Learner Guide — full detailed step-by-step instructions for all 11 hands-on labs.
- Lab code repository: <https://github.com/tertiarycourses/TGS-2020503207-AI-Vibe-Coding-for-Multi-Agents-System>
- Learner laptop with Python 3.11 or later, an IDE or terminal, and internet access.
- API keys for the model providers used in the labs (OpenAI and Google AI Studio).
- LMS/TMS portal: <https://lms-tms.tertiaryinfotech.com>

## Course Schedule

### Day 1 — Agent Foundations & Vibe Coding for Multi-Agent Systems

| Time        | Duration | Topic / Activity                                                                                                                                                                                                                                         |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–10:00  | 30 min   | Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory digital attendance (AM)                                                                                                                        |
| 10:00–11:00 | 60 min   | <b>Topic 1 — Modern Agent Foundations: what a modern AI agent is, the reason-act-observe loop, agent skills, memory, tools and the Model Context Protocol; the progression from single-agent to multi-agent collaboration (concepts + demonstration)</b> |
| 11:00–11:15 | 15 min   | Tea break                                                                                                                                                                                                                                                |
| 11:15–13:00 | 105 min  | Hands-on: Lab 1: Build Your First Tool-Calling Agent; Lab 2: Agent Memory and Reusable Agent Skills                                                                                                                                                      |
| 13:00–14:00 | 60 min   | Lunch break                                                                                                                                                                                                                                              |
| 14:00–15:00 | 60 min   | <b>Topic 2 — Vibe Coding for Multi-Agent Systems: the structured vibe coding workflow, context engineering, the vibe coding tool landscape and agent skills integration (concepts + demonstration). Digital attendance (PM)</b>                          |
| 15:00–16:30 | 90 min   | Hands-on: Lab 3: Vibe Code an Agent with a Structured                                                                                                                                                                                                    |

|             |        |                                                                                                                                                                    |
|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |        | Workflow; Lab 4: Context Engineering and a Reusable Coding Skill                                                                                                   |
| 16:30–16:45 | 15 min | Tea break                                                                                                                                                          |
| 16:45–17:30 | 45 min | <b>Topic 3 — Multi-Agent System Development with OpenAI Agents SDK: agents, runners, function tools and Pydantic structured outputs (concepts + demonstration)</b> |
| 17:30–18:45 | 75 min | Hands-on: Lab 5: First Agent with the OpenAI Agents SDK                                                                                                            |
| 18:45–19:00 | 15 min | Day 1 recap, Q&A and confirmation of digital attendance                                                                                                            |

*Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and two 15-minute tea breaks.*

## Day 2 — Agent SDKs, MCP, Sub-Agents & Assessment

| Time        | Duration | Topic / Activity                                                                                                                                                           |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:45   | 15 min   | Day 1 recap and mandatory digital attendance (AM)                                                                                                                          |
| 9:45–10:15  | 30 min   | <b>Topic 3 continued — supervisor routing, sub-agent delegation, handoffs and guardrails (concepts + demonstration)</b>                                                    |
| 10:15–11:15 | 60 min   | Hands-on: Lab 6: Supervisor Routing and Sub-Agent Delegation                                                                                                               |
| 11:15–11:30 | 15 min   | Tea break                                                                                                                                                                  |
| 11:30–12:30 | 60 min   | Hands-on: Lab 7: Deploy the Multi-Agent System with Streamlit                                                                                                              |
| 12:30–13:00 | 30 min   | <b>Topic 4 — Multi-Agent System Development with Gemini Agent SDK: agent design in the Google ADK and comparison with the OpenAI Agents SDK (concepts + demonstration)</b> |
| 13:00–14:00 | 60 min   | Lunch break                                                                                                                                                                |
| 14:00–15:30 | 90 min   | Hands-on: Lab 8: Build a Multi-Agent System with the Gemini Agent SDK; Lab 9: Deploy the Gemini Multi-Agent System with Streamlit. Digital attendance (PM)                 |
| 15:30–15:45 | 15 min   | Tea break                                                                                                                                                                  |
| 15:45–16:15 | 30 min   | <b>Topic 5 — MCP and Sub-Agents: MCP servers and tool orchestration in depth, and hierarchical sub-agent architecture (concepts + demonstration)</b>                       |
| 16:15–18:00 | 105 min  | Hands-on: Lab 10: Build and Connect an MCP Server; Lab                                                                                                                     |

|             |        |                                                                                                                                                                                      |
|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |        | 11: Hierarchical Sub-Agents and Context Isolation                                                                                                                                    |
| 18:00–18:30 | 30 min | Course revision and summary, course feedback and TRAQOM survey                                                                                                                       |
| 18:30–18:45 | 15 min | <b>Briefing for Assessment and Assessment digital attendance</b>                                                                                                                     |
| 18:45–19:00 | 15 min | <b>Final Assessment administration — the Written Assessment (WA, SAQ, 50 minutes) and Practical Performance (PP, 75 minutes) are conducted as scheduled by the assessment centre</b> |

*Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and two 15-minute tea breaks.*

## Lab Reference (aligned to the course outline)

| Topic                                                                  | Weighting | Labs                |
|------------------------------------------------------------------------|-----------|---------------------|
| <b>Topic 01: Modern Agent Foundations</b>                              | 20%       | Lab 1, Lab 2        |
| <b>Topic 02: Vibe Coding for Multi-Agent Systems</b>                   | 20%       | Lab 3, Lab 4        |
| <b>Topic 03: Multi-Agent System Development with OpenAI Agents SDK</b> | 25%       | Lab 5, Lab 6, Lab 7 |
| <b>Topic 04: Multi-Agent System Development with Gemini Agent SDK</b>  | 20%       | Lab 8, Lab 9        |
| <b>Topic 05: MCP and Sub-Agents</b>                                    | 15%       | Lab 10, Lab 11      |

## Learning Outcome to Lab Mapping

| Learning Outcome                                                                                                                                                                    | Topic                          | Assessed by   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------|
| LO1: Explain the components of a modern AI agent — skills, memory, tools and the Model Context Protocol (MCP) — and the progression from single-agent to multi-agent collaboration. | Topic 01 (Lab 1, Lab 2)        | WA (SAQ) + PP |
| LO2: Apply a structured vibe coding workflow with context engineering to specify, generate and verify agent code reliably.                                                          | Topic 02 (Lab 3, Lab 4)        | WA (SAQ) + PP |
| LO3: Build a collaborative multi-agent system with the OpenAI Agents SDK using structured outputs, tool calling and supervisor routing, and                                         | Topic 03 (Lab 5, Lab 6, Lab 7) | WA (SAQ) + PP |

|                                                                                                                     |                           |               |
|---------------------------------------------------------------------------------------------------------------------|---------------------------|---------------|
| deploy it with Streamlit.                                                                                           |                           |               |
| LO4: Build a collaborative multi-agent system with the Google Gemini Agent SDK and deploy it with Streamlit.        | Topic 04 (Lab 8, Lab 9)   | WA (SAQ) + PP |
| LO5: Orchestrate tools through MCP and design hierarchical sub-agent architectures for delegated, specialised work. | Topic 05 (Lab 10, Lab 11) | WA (SAQ) + PP |